

CV



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Profile

I have been coding in Python and **working with Artificial Intelligence/Machine Learning for five years**. I hold a **bachelor's degree in Mathematics** and a **master's degree in Data Science**. My expertise includes Statistics, Probability Theory, Linear Algebra, Calculus, Partial Differential Equations, and Ordinary Differential Equations, all integrated with Machine Learning applications. I have experience building Machine Learning projects using various frameworks and tools, including NumPy, SciPy, Pandas, Scikit-Learn, PyTorch, TensorFlow, FastAPI, GitHub, and Docker.

My projects primarily focus on improving model training time and reducing resource consumption. I believe that modern Machine Learning needs to be optimized for online streaming data, which demands high computational power and fast training. I have experience writing CUDA C to accelerate model training through parallel processing. Additionally, I research models that require fewer computational resources during the training process. I believe my skills can help you develop Machine Learning solutions that enhance your business.

Work Experience

- My Research: Strong-motion Earthquake Prediction Model using Convolutional Extreme Learning Machine [link](#)

- AI Engineer – Synapes Thailand | Remote Freelancer | Jan 2022 – Present
 - Cryptocurrency Portfolio Optimization – Developed optimization models for asset allocation, improving portfolio returns while reducing risk exposure.
 - Route Optimization – Designed AI-driven routing algorithms to minimize logistics costs and enhance operational efficiency.
 - Anomaly Detection using Machine Learning – Built anomaly detection systems for fraud prevention, improving security and reducing financial losses.
 - Churn Rate Analysis on Customer Data – Performed data cleaning and visualization to identify high-risk customers, helping improve retention strategies.
 - Predicting Default Debt Rate in Finance – Developed financial risk models to assess default probability, enabling businesses to manage risk more effectively.
 - Conducted training sessions on Machine Learning, PyTorch, and Reinforcement Learning for AIT Thailand. Designed and delivered hands-on workshops, guiding participants in implementing ML and RL models.
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Education

- Bachelor's Degree in Mathematics, Faculty of Science, Chiang Mai University (2014–2018)
 - Master's Degree in Data Science, Department of Engineering, Chiang Mai University (2021–Present)
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Skills

- Programming Languages:
 - Python: Python Data Analysis, Machine learning, AI development,
 - Julia: Mathemtic Models, Simulation,
 - C: Basic programing,
 - CUDA C: Optimization,
 - NetLogo: Agent-Based modeling
- Frameworks & Libraries:
 - NumPy,
 - SciPy,
 - Pandas,
 - Scikit-Learn,
 - PyTorch,
 - TensorFlow,
 - Z3,
 - NetworkX,
 - JupyterLab,
 - FastAPI
- Tools & Technologies:
 - Docker,
 - Git,
 - Streamlit,
 - Gradio
- Models & Techniques:
 - Extreme Learning Machine,
 - Reservoir Computing,
 - Reinforcement Learning,

- Agent-Based Modeling,
 - Attention Mechanism,
 - Transformers,
 - Dendritic Gated Networks,
 - Hyperdimensional Computing,
 - Conformal Prediction,
 - Extreme Value Theory,
 - Portfolio Optimization,
 - Particle Swarm Optimization,
 - Genetic Algorithm,
 - Cryptography,
 - Quantum Computing,
 - Monte Carlo Simulation
 - Large Language Model
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Personal Projects

- My medium blog: I write many articles about Probability theory, Mathematics and AI
 - Using Extreme Value Theory to analyze the worst scenario in the Crypto Market [link](#)
 - Trading Strategy Assessment By The Bootstrap Method. [link](#)
 - Fermat Factorization Algorithm can break poor RSA encryption. [link](#)
 - Let's code Lattice-Based Encryption: The post-quantum encryption. [link](#)
 - Deep learning in the Matrix [link](#)
 - Extreme Learning Machine (ELM) is the speed-up learning method for Artificial Neural Network [link](#)
 - Let's deploy your deep learning model with Gradio. [link](#)
 - etc.
- Academic Papers
 - A Product of Two Primes with Difference 2 [link](#)
- Conferences
 - Strong-motion Earthquake Prediction Model using Convolutional Extreme Learning Machine [link](#)
 - Auto Encoder for Anomaly Detection in the Cryptocurrency Market Using On-Chain Data [link](#)
- GitHub Project
 - Complexity Science Project [link](#)
 - Quantum Tutorial [link](#)
 - Implement ML using FastAPI [link](#)
 - Basic PyTorch Tutorial [link](#)
 - TensorFlow tutorial [link](#)
 - Air Pollution Prediction using GNN [link](#)
 - Genetic Algorithm for Mathematics [link](#)
 - Z3 optimization [link](#)
 - Topic Clustering with LDA [link](#)
 - Bird Song Detection using Deep Neural Network [link](#)
 - Time Table Management using GA [link](#)

Certifications

- Super AI Engineer Season 2
- Automated Reasoning: satisfiability [link](#)
- Neural Networks and Deep Learning [link](#)
- Biology Meets Programming: Bioinformatics for Beginners [link](#)
- Introduction to Agent-Based Modeling [link](#)
- Nonlinear Dynamics: Mathematical and Computational Approaches [link](#)
- Introduction to Dynamical Systems and Chaos [link](#)
- Fractals and Scaling [link](#)